

360Tools.me — Existing Website Refactor, SEO & Performance Optimization

ROLE

Act as a senior frontend engineer, technical SEO specialist, performance engineer, and UX engineer.

You are working on an **existing production website: 360tools.me**.

Do NOT rebuild the website from scratch.

Your job is to **inspect the existing codebase first**, understand how the current website and tools work, and then refactor it into a cleaner, faster, SEO-friendly static architecture while preserving all existing functionality.

1. PRIMARY OBJECTIVES

Refactor the existing 360Tools website to achieve:

- Better technical SEO
- Better URL/site architecture
- Better Google crawling and indexing
- Better Core Web Vitals
- Better PageSpeed performance
- Lower JavaScript payload
- Lower CSS payload
- Faster first load
- Better internal linking
- Better category organization
- Better mobile UX
- Better accessibility
- Better maintainability
- Preserve all existing tools and functionality
- Preserve the existing visual identity unless a UI improvement is clearly necessary

The website should remain lightweight.

Do NOT introduce React, Vue, Angular, Next.js, or another frontend framework.

Use:

- HTML5

- CSS3
 - Vanilla JavaScript
 - Existing libraries only when genuinely required
-

2. FIRST STEP — AUDIT THE EXISTING PROJECT

Before modifying anything, inspect the entire project.

Identify:

- Existing HTML pages
- Existing tool pages
- Existing categories
- Existing CSS
- Existing JavaScript
- Existing images
- Existing fonts
- Existing third-party libraries
- Existing analytics
- Existing advertising scripts
- Existing navigation
- Existing footer
- Existing SEO metadata
- Existing canonical URLs
- Existing sitemap.xml
- Existing robots.txt
- Existing redirects/configuration
- Duplicate CSS/JS
- Unused CSS/JS
- Large assets
- Render-blocking resources
- Pages loading unnecessary JavaScript
- Broken links
- Missing alt text
- Missing width/height attributes
- Duplicate titles/descriptions
- Missing H1 tags
- Poor heading hierarchy

Do not blindly modify files.

First understand the current architecture and dependencies.

3. TARGET SITE ARCHITECTURE

Refactor the website toward this structure:

360tools.me/

└── index.html

└── calculators/ | └── mortgage-calculator.html | └── mortgage-payment-calculator.html |
└── property-tax-calculator.html | └── sales-tax-calculator.html | └── income-tax-calculator.html
| └── tip-calculator.html | └── 401k-calculator.html | └── compound-interest-calculator.html |
└── loan-payment-calculator.html | └── credit-card-payoff-calculator.html | └── hourly-to-salary-
calculator.html | └── salary-to-hourly-calculator.html | └── overtime-calculator.html

└── image-tools/ | └── image-compressor.html | └── image-resizer.html | └── image-
converter.html | └── jpg-to-png.html | └── png-to-jpg.html | └── background-remover.html

└── pdf-tools/ | └── pdf-to-jpg.html | └── jpg-to-pdf.html | └── pdf-compressor.html

└── text-tools/ | └── word-counter.html | └── character-counter.html | └── case-
converter.html | └── text-cleaner.html

└── audio-tools/ | └── existing audio tools

└── developer-tools/ | └── existing developer tools

└── finance/ | └── existing finance tools

└── assets/ | └── css/ | | └── main.min.css | | └── tool.min.css | | | └── js/ | |
└── main.min.js | | └── mortgage-calculator.min.js | | └── image-compressor.min.js | | └──
pdf-tools.min.js | | └── ... | | | └── images/ | └── icons/

└── robots.txt | └── sitemap.xml | └── manifest.webmanifest

IMPORTANT:

Do not move a page into a category merely because it sounds appropriate.

Inspect the existing tools and place them into logical categories.

Do not delete any existing working tool.

4. URL MIGRATION

If existing pages currently look like:

`/mortgage-calculator.html`

and the new architecture requires:

`/calculators/mortgage-calculator.html`

then:

1. Create the new URL.
2. Preserve all existing functionality.
3. Update every internal link.
4. Update canonical URLs.
5. Update sitemap.xml.
6. Update breadcrumbs.
7. Update navigation.
8. Add a proper 301 redirect from the old URL to the new URL where the hosting environment supports redirects.

Do NOT create redirect loops.

Do NOT leave duplicate versions of the same page indexable.

Use the new URL as the canonical URL.

IMPORTANT:

The website is already connected to Google Search Console.

Do not assume that Search Console pages need to be manually recreated.

The migration should be technically clean so Google can discover the new URLs through:

- 301 redirects
 - sitemap.xml
 - internal links
 - canonical tags
-

5. SEO FOR EVERY TOOL PAGE

Every important tool page should have unique SEO metadata.

Example:

```
<title>Mortgage Calculator - Calculate Monthly Payments | 360Tools</title>
```

```
<meta name="description" content="Use our free mortgage calculator to estimate monthly payments, principal, interest and total mortgage costs.">
```

```
<link rel="canonical" href="https://360tools.me/calculators/mortgage-calculator.html">
```

Do NOT copy the same title and description across every page.

Each page should have:

- Unique title
- Unique meta description
- One clear H1
- Proper H2/H3 hierarchy
- Canonical URL
- Relevant Open Graph metadata
- Relevant Twitter metadata where appropriate
- Descriptive image alt text
- Breadcrumbs where appropriate

6. TOOL PAGE CONTENT STRUCTURE

Do not make tool pages just a calculator with a button.

Where appropriate, structure pages like:

H1 — Mortgage Calculator

Short introduction

Calculator UI

Results

How to Use This Calculator

How Mortgage Payments Are Calculated

Formula

Example Calculation

Frequently Asked Questions

Related Calculators

Important notes/disclaimers where appropriate

The actual calculator should remain prominent.

Do not add useless AI-generated filler.

Content must be:

- Useful
- Concise
- Human-readable
- Relevant to the tool
- Unique
- Accurate

7. INTERNAL LINKING

Build a strong internal-linking system.

Homepage should link to major categories.

Category pages should link to individual tools.

Individual tools should link to relevant related tools.

Example:

Mortgage Calculator

Related Tools:

- Mortgage Payment Calculator
- Property Tax Calculator
- Loan Payment Calculator

- Compound Interest Calculator
- Salary Calculator

Use normal HTML links.

Do not use JavaScript navigation for important SEO links.

8. CATEGORY PAGES

Create or improve category landing pages where appropriate.

For example:

/calculators/

/image-tools/

/pdf-tools/

/text-tools/

/audio-tools/

Category pages should contain:

- H1
- Short useful introduction
- Tool cards
- Links to all relevant tools
- Short category description
- Good internal linking
- SEO metadata

Do not make category pages thin or empty.

9. PERFORMANCE OPTIMIZATION

The website should be optimized specifically for PageSpeed Insights and Core Web Vitals.

Target:

Performance: 90–100

Accessibility: 95+

Best Practices: 95+

SEO: 95+

Aim for:

LCP < 2.5 seconds

INP < 200ms

CLS < 0.1

Optimize for real users, not just the score.

10. JAVASCRIPT ARCHITECTURE

This is extremely important.

DO NOT load every tool's JavaScript on every page.

Bad:

```
<script src="/assets/js/all-tools.js"></script>
```

if it contains code for hundreds of unrelated tools.

Instead use page-specific JavaScript.

Example:

Mortgage page:

```
<script src="/assets/js/main.min.js" defer></script> <script src="/assets/js/mortgage-calculator.min.js" defer></script>
```

Image compressor:

```
<script src="/assets/js/main.min.js" defer></script> <script src="/assets/js/image-compressor.min.js" defer></script>
```

PDF tool:

```
<script src="/assets/js/main.min.js" defer></script> <script src="/assets/js/pdf-tools.min.js" defer></script>
```

Use defer for non-critical scripts.

Do not load unnecessary libraries.

11. CSS OPTIMIZATION

Create a lightweight shared CSS architecture.

Example:

/assets/css/main.min.css

contains:

- Reset/base styles
- Typography
- Header
- Navigation
- Footer
- Buttons
- Cards
- Forms
- Responsive utilities

Then use tool-specific CSS only when required.

Do not load a huge CSS file containing styles for every possible tool if the current page only needs a small subset.

Remove:

- Duplicate CSS
- Unused CSS
- Old styles
- Test styles
- Dead classes

Preserve the current 360Tools visual identity.

12. IMAGES

Optimize every image.

Prefer:

- SVG for logos/icons/simple graphics
- WebP
- AVIF where appropriate

Avoid unnecessarily large PNG/JPEG files.

Add:

- width
- height
- alt

to images where appropriate.

Use lazy loading for below-the-fold images.

Do NOT lazy-load the primary LCP image.

Example:

```

```

13. FONTS

Reduce font loading.

Do not load unnecessary font families or weights.

Prefer:

- System fonts

or a minimal font setup.

Avoid loading 8–10 font weights.

Do not use external font requests unless necessary.

14. THIRD-PARTY SCRIPTS

Audit all third-party scripts.

Potential examples:

- Google Analytics
- Google AdSense
- Analytics
- Social widgets
- Chat widgets
- External embeds

Do not remove required monetization/tracking functionality without checking its purpose.

However:

- Defer non-critical scripts where safe
- Do not load unnecessary widgets
- Do not load third-party scripts on pages where they are not needed

The goal is to keep the site fast while preserving monetization.

15. CORE WEB VITALS

Specifically inspect and improve:

LCP

Find the current LCP element.

Optimize:

- Hero images
- Fonts
- Critical CSS
- HTML delivery
- Server response
- Above-the-fold content

CLS

Reserve dimensions for:

- Images
- Ads
- Dynamic content
- UI components

Do not allow content to jump when resources load.

INP

Reduce:

- Large JavaScript execution
- Long event handlers
- Unnecessary DOM manipulation
- Heavy animations

Keep calculator interactions fast.

16. ACCESSIBILITY

Improve:

- Semantic HTML
- Form labels
- Button labels
- Keyboard navigation
- Focus states
- Color contrast
- ARIA only when actually necessary
- Alt text

Every calculator input should have an accessible label.

Example:

```
<label for="loanAmount">Loan Amount</label> <input id="loanAmount" ...>
```

Do not use placeholder text as the only label.

17. STRUCTURED DATA

Use Schema.org structured data only where it genuinely applies.

Consider:

- WebSite
- BreadcrumbList
- WebApplication where appropriate

Do NOT add fake reviews.

Do NOT add fake ratings.

Do NOT add misleading structured data.

Validate structured data after implementation.

18. SITEMAP.XML

Generate/update sitemap.xml.

Include only:

- Canonical
- Indexable
- Working URLs

Do not include:

- Redirect URLs
- 404 pages
- Duplicate URLs
- Noindex pages
- Test pages

Example:

<https://360tools.me/>

<https://360tools.me/calculators/mortgage-calculator.html>

<https://360tools.me/calculators/compound-interest-calculator.html>

etc.

19. ROBOTS.TXT

Use a simple configuration unless the existing site has a specific reason for something else:

User-agent: * Allow: /

Sitemap: <https://360tools.me/sitemap.xml>

Do not accidentally block CSS, JS, images or important pages.

20. CANONICALIZATION

Every indexable tool page should have a self-referencing canonical URL.

Example:

```
<link rel="canonical" href="https://360tools.me/calculators/mortgage-calculator.html">
```

Do not create multiple canonical URLs for the same content.

21. MOBILE-FIRST

The entire website must work properly on:

- Mobile
- Tablet
- Desktop

Test:

320px 375px 390px 768px 1024px 1440px+

Calculator controls must be comfortable to use on mobile.

Avoid horizontal scrolling.

22. DESIGN REQUIREMENTS

Keep 360Tools looking like a professional real product.

Do NOT redesign it into:

- Generic AI landing page
- Excessive gradients
- Excessive glassmorphism
- Huge animations
- Unnecessary 3D
- Excessive shadows
- Heavy particle effects

Prioritize:

- Clean
- Premium
- Fast
- Human-designed
- Functional
- Trustworthy

The tool should be the focus.

23. ADS / MONETIZATION

The site is intended for monetization through advertising.

Do not destroy existing AdSense/advertising integration.

However:

- Reserve ad space to reduce CLS
- Avoid placing ads above critical calculator functionality
- Do not overload pages with ads
- Keep the main tool immediately usable
- Avoid intrusive popups

The user experience must remain strong.

24. SEO-FRIENDLY HTML

Prefer semantic HTML:

```
<header> <nav> <main> <section> <article> <footer>
```

Use normal anchor links:

```
<a href="/calculators/mortgage-calculator.html"> Mortgage Calculator </a>
```

Avoid making SEO navigation dependent entirely on JavaScript.

25. 404 PAGE

Create/improve:

/404.html

It should:

- Look like the main website
- Explain that the page doesn't exist
- Link back to homepage
- Link to major categories
- Have useful navigation

Do not automatically redirect every 404 to the homepage.

26. PERFORMANCE BUDGET

Aim to keep initial page resources small.

For ordinary calculator pages, target approximately:

HTML: <100 KB

CSS: <100 KB

Initial JS: <100 KB where practical

Images: minimal

Avoid loading libraries that aren't required.

Tool-specific heavy libraries such as PDF/image processing libraries should only load on pages that actually require them.

27. IMPORTANT — PRESERVE FUNCTIONALITY

Before changing a tool:

Understand how it currently works.

Do not break:

- Calculations
- File processing
- Downloads
- Uploads
- Converters
- Compressors
- Forms
- Validation
- Local browser processing
- Existing APIs
- Existing integrations

After refactoring each tool, test it.

28. DO NOT DELETE CONTENT

Do not delete existing useful SEO content.

Do not remove existing tools.

Do not remove pages merely because they are currently poorly optimized.

Improve them.

If duplicate pages exist, identify them first and report them before deleting anything.

29. BUILD A MIGRATION REPORT

After completing the refactor, create:

SEO-MIGRATION-REPORT.md

Include:

1. Old URL
2. New URL
3. Redirect required?
4. Canonical URL
5. Sitemap status
6. Internal links updated?
7. Metadata updated?
8. Structured data status

Example:

Old URL	New URL	Redirect	Canonical
/mortgage-calculator.html	/calculators/mortgage-calculator.html	301	New URL

30. CREATE PERFORMANCE REPORT

Create:

PERFORMANCE-REPORT.md

Include:

- Largest files
- Largest JS files
- Largest CSS files
- Large images
- Third-party scripts
- Potential render-blocking resources
- Unused assets
- Optimization performed
- Remaining recommendations

31. CREATE SEO REPORT

Create:

SEO-AUDIT.md

Include:

- Number of pages
 - Number of indexable pages
 - Missing titles
 - Duplicate titles
 - Missing descriptions
 - Duplicate descriptions
 - Missing H1
 - Broken links
 - Missing canonical tags
 - Sitemap status
 - Robots.txt status
 - Internal linking improvements
 - Redirects created
 - Structured data implemented
-

32. IMPORTANT IMPLEMENTATION RULE

Do this in phases.

PHASE 1: Audit existing website.

PHASE 2: Create new folder architecture.

PHASE 3: Migrate pages.

PHASE 4: Update internal links.

PHASE 5: Implement redirects.

PHASE 6: SEO metadata/canonical/breadcrumbs.

PHASE 7: Sitemap + robots.txt.

PHASE 8: JavaScript/CSS optimization.

PHASE 9: Image/font optimization.

PHASE 10: Accessibility.

PHASE 11: Performance testing.

PHASE 12: Final SEO audit.

Do not perform a destructive mass rewrite without first understanding the existing code.

33. FINAL ACCEPTANCE CHECKLIST

Before declaring the work complete, verify:

- ☐ Homepage works
- ☐ Every existing tool still works
- ☐ New category structure works
- ☐ All internal links work
- ☐ No broken links
- ☐ No redirect loops
- ☐ Old URLs redirect correctly where applicable
- ☐ New URLs have canonical tags
- ☐ Sitemap contains new canonical URLs
- ☐ Robots.txt works
- ☐ Every important page has unique title
- ☐ Every important page has unique description
- ☐ Every important page has one H1
- ☐ Images have appropriate alt text
- ☐ Images have dimensions where appropriate

- ☐ Mobile layout works
 - ☐ Desktop layout works
 - ☐ JavaScript is page-specific
 - ☐ CSS is optimized
 - ☐ Fonts are optimized
 - ☐ Third-party scripts are audited
 - ☐ CLS is minimized
 - ☐ LCP is optimized
 - ☐ INP is optimized
 - ☐ Accessibility improved
 - ☐ 404 page works
 - ☐ SEO-AUDIT.md created
 - ☐ PERFORMANCE-REPORT.md created
 - ☐ SEO-MIGRATION-REPORT.md created
-

MOST IMPORTANT INSTRUCTION

This is an **existing website with existing users, existing Google indexing, existing tools and existing functionality**.

Therefore:

DO NOT rebuild blindly.

DO NOT delete working tools.

DO NOT change URLs without redirects.

DO NOT create duplicate indexable pages.

DO NOT sacrifice functionality for PageSpeed.

DO NOT add unnecessary frameworks.

DO NOT add fake SEO content.

DO NOT add keyword stuffing.

DO NOT make the website look AI-generated.

Build a fast, clean, professional, human-designed static tools website that is technically strong for Google Search and pleasant for real users.

At the end, provide a concise summary of:

1. What was changed
2. What URLs changed
3. What redirects were created
4. What SEO improvements were made
5. What performance improvements were made
6. Any remaining issues
7. Files created/modified